

# Criterion III.4

Data Protection Requirements, Training and Regulatory Obligations

## GOLDBOD TAS

### Criterion III.4 Response

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Smart Innovations Ghana Limited | Sylvester Akpah

GR/GGB/TS/NCT/042/2026 | April 2026

# III.4

## Criterion Data Protection Requirements, Training and Regulatory Obligations

### Data Protection Act, 2012 (Act 843) Compliance

The GoldBod TAS is designed and implemented in full compliance with the Data Protection Act, 2012 (Act 843) of Ghana. Every data handling decision in the system architecture maps to a specific obligation under the Act:

#### Act 843 Principles — Implementation Controls

- Purpose limitation: actor PII (name, NIA number, licence details) collected exclusively for gold chain-of-custody tracking — no secondary use without GoldBod authorisation
- Data minimisation: field actors provide only data required for their specific role. Miners cannot access buyer commercial data; buyers cannot see other buyers' transactions
- Storage limitation: custody records retained indefinitely (AML Act 1044 mandate); session tokens and temporary data deleted on expiry via automated retention policies
- Access controls: role-based access control ensures each user accesses only data relevant to their actor type and assigned responsibilities — enforced at API level, not just UI level
- Data subject rights: actor records updatable by authorised GoldBod administrators. Deactivation workflow preserves audit history while removing operational access
- Security measures: TLS 1.3 in transit, PostgreSQL encryption at rest, JWT token expiry, bcrypt password hashing, Ed25519 device key authentication

### Anti-Money Laundering Act, 2020 (Act 1044) Compliance

|                              |                                                                                                                                                                                       |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actor identity verification  | Every registered actor linked to Ghana Card (NIA) number. KYC status: PENDING / VERIFIED / FLAGGED / SUSPENDED. Licence expiry enforced — expired-licence actors cannot transact.     |
| Beneficial ownership         | Corporate entity actors require beneficial ownership declaration at registration — addresses Act 1044 Section on corporate transparency and ultimate beneficial owner identification. |
| Transaction record retention | All CustodyEvents recorded with immutable timestamps, actor identities, GPS coordinates, weights. Retention: indefinite per Act 1044 requirements.                                    |
| STR / SAR capability         | Suspicious Transaction Report workflow in compliance module. goAML XML generation for FIC submission. Compliance officer dashboard with mandatory disposition justification.          |
| Sanctions screening          | Actor onboarding triggers screening against UN Security Council, OFAC SDN, EU, UK HMT, and Ghana FIC watchlists. Every check logged permanently.                                      |
| AML rules engine             | Velocity monitoring, border-proximity flagging, structuring detection, grade anomaly detection, counterparty network analysis, dormancy-reactivation pattern detection.               |

### Regulatory Obligations — TOR Section Mapping

|                                                 |                                                                                                                                                                            |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TOR 2.2.3 — Identity &amp; Due Diligence</b> | Implemented: verified actor registry, all 11 licence categories, KYC/AML status, Ghana Card NIA integration, KYC/CFT compliance per GoldBod Act 1044 obligations.          |
| <b>TOR 2.2.4 — Compliance &amp; Reporting</b>   | Implemented: AML/CFT monitoring framework, compliance dashboard, STR/SAR generation, FIC API integration pathway, FATF-aligned reporting.                                  |
| <b>TOR 2.2.1 — Origin &amp; Custody</b>         | Implemented: full chain of custody from miner through smelting to GoldBod intake. Trace ID system. React Flow audit graph with full provenance tree.                       |
| <b>TOR 3.3 — Data Governance &amp; IP</b>       | Implemented: all data generated under TAS is GoldBod's property. Source code in GoldBod-accessible repository. No proprietary lock-in. Full escrow arrangements available. |
| <b>TOR 2.2.5 — System Sustainability</b>        | Implemented: PM2 uptime management, offline mobile app for field use, defined DR and national-scale migration path. 18-month delivery plan with GoldBod handover.          |
| <b>TOR 3.1 — Blockchain Responsibilities</b>    | Implemented (Phase 2): Hyperledger Fabric 2.5 permissioned ledger, GoldBod MSP CA, chaincode for actor onboarding, custody transfer, compliance recording.                 |

## Training Programme

Smart Innovations Ghana Limited will deliver a structured, tiered training programme. Materials are prepared for mixed digital literacy levels and delivered in English, Twi, Ga, and Hausa:

|                                      |                                                                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GoldBod Officers</b>              | 3-day administrator training: system configuration, licence management, compliance dashboard, STR workflow, Hyperledger Explorer, user management.             |
| <b>Compliance Officers</b>           | 2-day focused training: AML/CFT rules engine, flag review and disposition, STR/SAR filing, FIC integration, regulatory report exports.                         |
| <b>Smelting / Kitchen Operators</b>  | 2-day operational training: lot intake, assay data entry, amalgamation workflow, batch management, Explorer dashboard read access.                             |
| <b>Field Actors (Miners, Buyers)</b> | 1-day field training: mobile app operation, QR/NFC scanning, offline mode, lot creation, custody transfer. Train-the-trainer: 5 regional champions per region. |
| <b>IT / System Administrators</b>    | 4-day technical handover: infrastructure management, database administration, deployment pipeline, incident response, chaincode upgrade procedures.            |
| <b>Ongoing support</b>               | Help desk: SMS + WhatsApp (Tier 1), phone (Tier 2), engineering escalation (Tier 3). Hours: 08:00–18:00 GMT. 90-day post-go-live warranty.                     |

### EVIDENCE PROVIDED

- ✓ Full Act 843 compliance: purpose limitation, data minimisation, access controls, retention policies — all enforced at architecture level
- ✓ Act 1044 controls: actor identity (NIA), beneficial ownership, immutable transaction records, STR/SAR workflow, sanctions screening
- ✓ All 6 TOR regulatory sections (2.2.1, 2.2.3, 2.2.4, 2.2.5, 3.1, 3.3) mapped to implemented system features
- ✓ Role-based access control with JWT — actor-type data boundaries enforced at API level
- ✓ 5-tier structured training programme: GoldBod officers, compliance, kitchen operators, field actors, IT staff
- ✓ 3-tier help desk support + 90-day post-go-live warranty period